

THC block-encoding Toffoli comparison: four constructions

Parameters: $N_{\text{up}} = 4$, $N_{\text{down}} = 22$, N_{IP} (central bond) = 256
 $N_k = k^3$ for k in $[1, 2, 3, 4, 5, 6]$ (so N_k in $[1, 8, 27, 64, 125, 216]$)
phase_bitsize $b = 32$

All four encode the periodic THC tensor (arXiv:2601.16379 Eq. 9):

1. Direct: orientation flipped -- inputs $(\mu, k_\mu), (\lambda, k_\lambda)$, outputs the other two. $|Q\rangle$ uniform LCU, two X reflections, two momentum modular adds $k \pm Q$, central DIAGONAL kernel sum $W^{\{IJ\}}|QIJ\rangle\langle QIJ|$ (QROAM $\rightarrow$ Ry $\rightarrow$ QROAM †), then the two X † .

Exchange variants share bra-pair X reflections; they differ in the central W^q :

2. Exchange US (Unitary Synthesis): central W^q via the SVD block-unitary interferometer (default central encoder).
3. Exchange Frobenius (use_fro_BE=True): central W^q via the Clader-Frobenius block-matrix scheme (BlockDiagonalClassicalMatrixBlockEncoding, $U_A = U_L^\dagger U_R$).
4. Exchange Reflection (central_via_reflection=True): central W^q via the Householder reflection isometry (ReflectionRectangularBlockEncoding, N_{IP} refl).

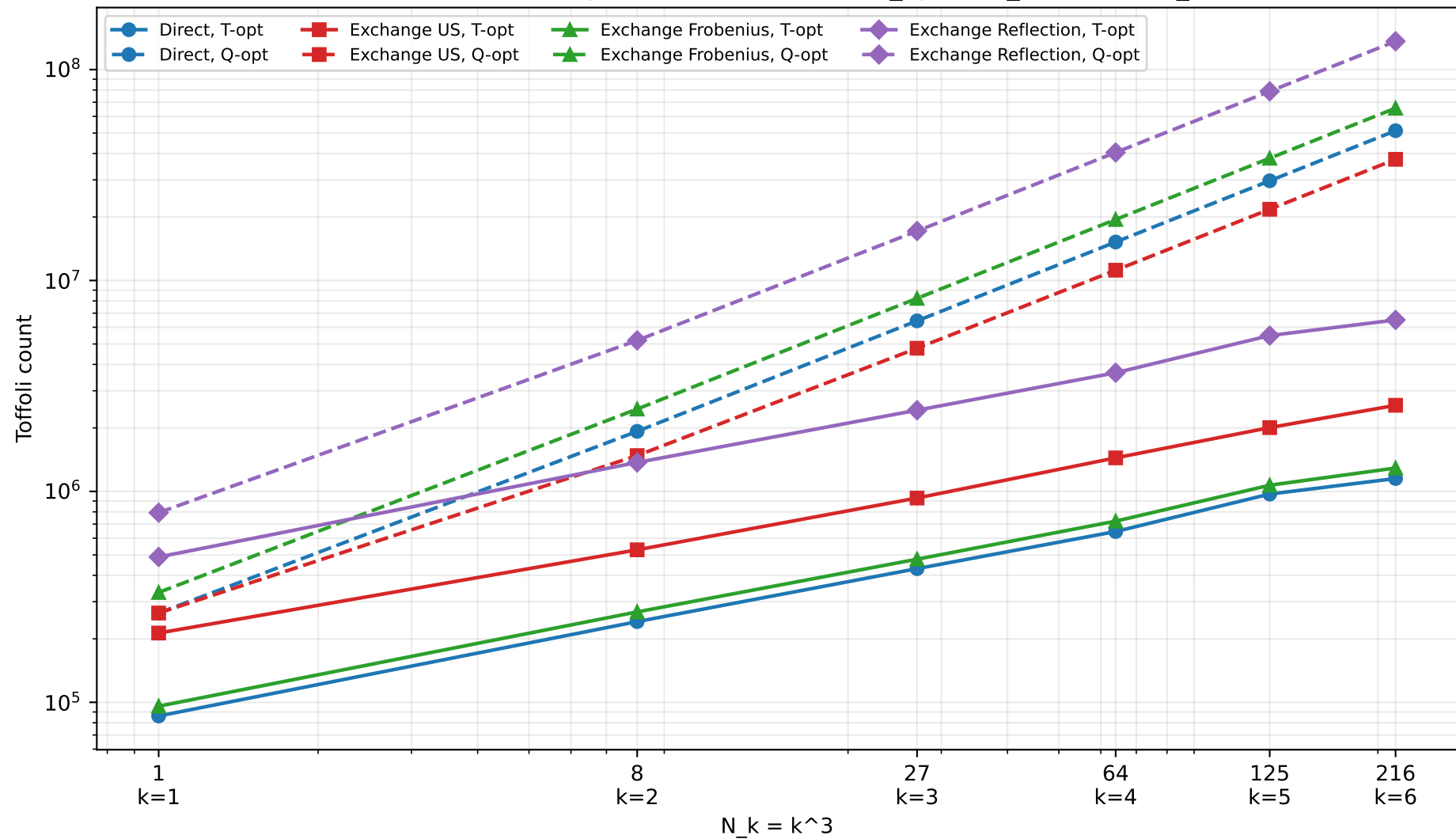
Tuning policies (per curve):

T-opt: every QROAM log_block_sizes at the closed-form Toffoli optimum.

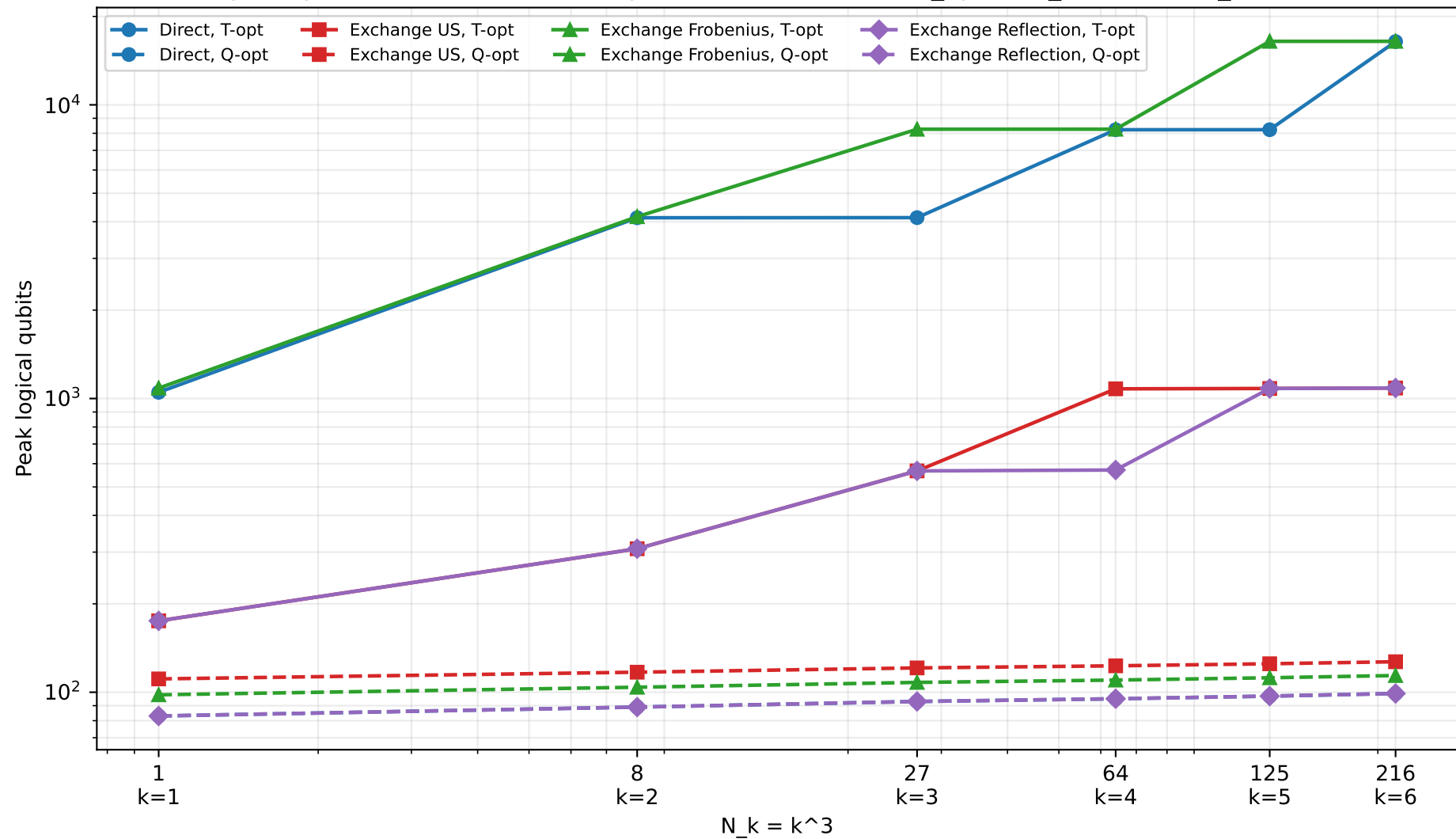
Q-opt: no QROAM blocking (log_block_sizes = 0) $\rightarrow$ minimal qubits.

All Toffoli / qubit counts come from Qualtran's QECGatesCost / QubitCount walking each bloq's build_call_graph (no analytic formulas).

Toffoli count: THC block encoding, four constructions (N_up=4, N_down=22, N_IP=256, b=32)



Peak logical qubits: THC block encoding, four constructions ($N_{\text{up}}=4$, $N_{\text{down}}=22$, $N_{\text{IP}}=256$, $b=32$)



Toffoli count: THC block encoding, four constructions

construction	N_k=1	N_k=8	N_k=27	N_k=64	N_k=125	N_k=216
Direct, T-opt	86,130	241,310	430,334	645,164	971,550	1,150,724
Direct, Q-opt	264,923	1,926,447	6,436,319	15,218,613	29,697,615	51,297,365
Exchange US, T-opt	213,032	528,414	930,190	1,441,878	2,006,410	2,560,306
Exchange US, Q-opt	264,973	1,478,501	4,772,377	11,186,673	21,761,677	37,537,429
Exchange Frobenius,	95,863	268,303	477,031	722,596	1,069,631	1,290,669
Exchange Frobenius,	331,734	2,457,410	8,227,119	19,462,774	37,986,505	65,620,194
Exchange Reflection,	488,796	1,371,532	2,426,160	3,651,360	5,480,308	6,501,628
Exchange Reflection,	792,484	5,199,356	17,160,880	40,454,280	78,856,996	136,146,220

Peak logical qubits: THC block encoding, four constructions

construction	N_k=1	N_k=8	N_k=27	N_k=64	N_k=125	N_k=216
Direct, T-opt	1,050	4,126	4,130	8,227	8,229	16,422
Direct, Q-opt	83	89	93	95	97	99
Exchange US, T-opt	175	308	567	1,079	1,082	1,086
Exchange US, Q-opt	111	117	121	123	125	127
Exchange Frobenius,	1,085	4,161	8,260	8,262	16,455	16,457
Exchange Frobenius,	98	104	108	110	112	114
Exchange Reflection,	175	308	567	571	1,082	1,086
Exchange Reflection,	83	89	93	95	97	99